Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

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Abstract

Synthetic multi-asset equity paths must preserve both the per-asset return distributions and their relationship with the market. A common construction inserts a full-return generator draw directly into a single-index model, but this double counts the market variance and distorts the composed marginal. We derived a closed-form scale correction for this problem and evaluated it on 423 non-market assets in a 424-asset US equity and exchange-traded-fund universe. The correction set each synthetic asset's second moment to its empirical target, recovered the calibrated factor loading with low error, and preserved heavy-tailed marginals, all without a separate fit to the regression residuals. Residual models fitted directly to each asset passed the marginal goodness-of-fit tests more often, but only at the cost of a per-asset calibration stage; both approaches held one-day Value-at-Risk exceedances near their nominal level. The per-asset input was a hidden Markov model with Laplace-quantile states and Student-t emissions, whose jump-duration variant used Poisson episodes to lengthen visits to the tail states. On a decade of daily SPY data, this variant lowered the absolute-return autocorrelation error relative to its no-jump counterpart while keeping strong distributional fit, a tradeoff we used to select it as the marginal component rather than a claim that it dominated every baseline. The multi-asset construction still omits cross-asset residual covariance, so independent residuals can overstate diversification gains under daily rebalancing. Code and the per-asset fitted marginals are released with the paper.

Keywords: hidden Markov model; jump-duration mechanism; single-index model; synthetic financial data; variance correction; Value-at-Risk coverage check; volatility clustering; heavy-tailed marginals.

1 Introduction

Synthetic financial data often fail to reproduce the statistical properties of real returns. These paths are used to stress-test risk models against unobserved market scenarios, validate portfolio optimization algorithms, and Monte Carlo backtest allocators across many independent scenarios [1, 2]. Empirical equity excess growth rates exhibit three statistical regularities a generative model must reproduce: a heavy-tailed marginal distribution, weak autocorrelation of raw growth rates, and persistent volatility

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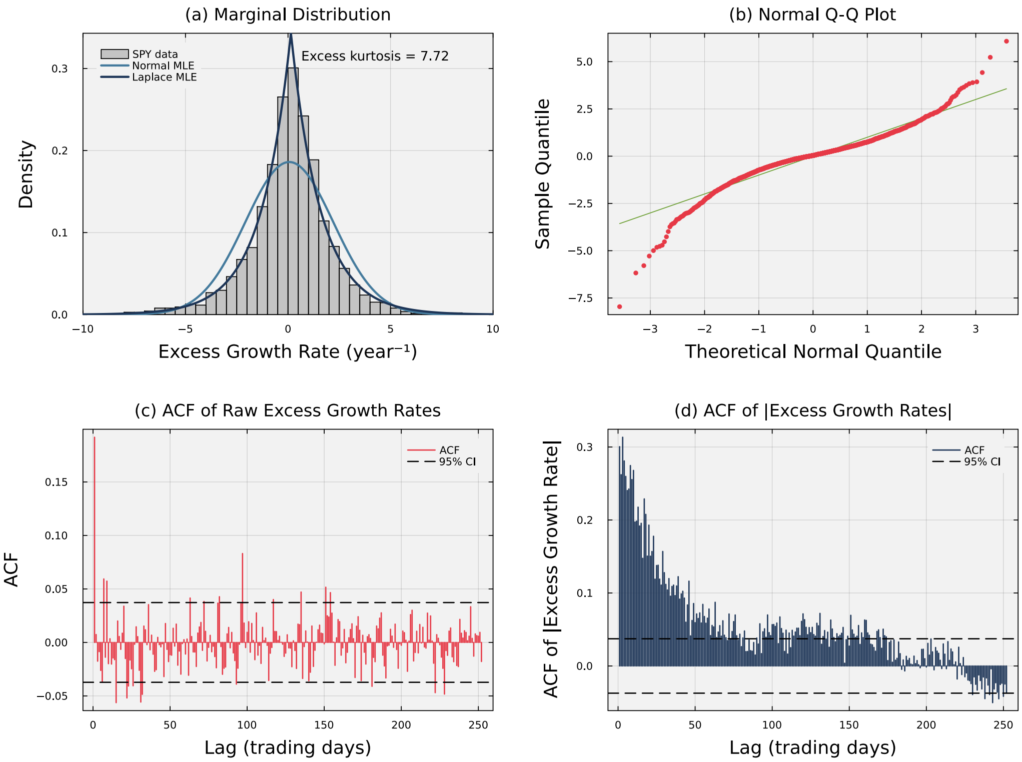

Figure 1: **Empirical stylized facts of SPY daily excess growth rates (**2014–2024**).** Panel (a) shows the leptokurtic marginal density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) shows departure from the diagonal in a normal quantile-quantile plot at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates, which stays near zero across all lags and is consistent with the efficient markets hypothesis. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly across the reported lags.

clustering, in which large changes are followed by large changes and small changes by small changes [3–5]. All three patterns are visible in SPY daily excess growth rates from 2014–2024 (Fig. 1). The empirical density is sharply concentrated near zero but places substantially more probability in the tails than a Gaussian model, as shown by the density and quantile-quantile comparisons (Fig. 1a and Fig. 1b). Raw growth rates have little serial correlation (Fig. 1c), whereas the positive, slowly decaying autocorrelation of their absolute values (Fig. 1d) shows that their magnitudes remain temporally dependent. These regularities, called *stylized facts*, demonstrate why matching the marginal distribution alone is insufficient: synthetic financial data must also preserve the distinct dependence structures of return levels and return magnitudes.

Existing generators typically reproduce some of these regularities but not others. The generalized autoregressive conditional heteroskedasticity (GARCH) family [6, 7] captures volatility clustering but requires fat-tailed innovations to reproduce heavy tails. Stochastic-volatility and jump-diffusion models [8–10] produce heavy tails, but jump effects do not persist. Deep generative models [11–13] can learn complex marginals but often miss the absolute-return autocorrelation associated with volatility clustering. Standard hidden Markov models (HMMs) [14] provide a regime-switching structure but return to central regimes too quickly to dwell in tail states long enough to reproduce the empirical clustering profile [15, 16]. We therefore sought a model that retained distributional fit, reduced absolute-return autocorrelation error, and could be fitted separately across a large asset universe.

In this study, we developed a multi-asset equity simulation framework that combines separately fitted hidden Markov marginals with a variance-corrected single-index composition. Each marginal model used a Laplace-quantile state partition, Student-t emissions, and direct transition counting, while the jump-duration variant (HMM-WJ) added Poisson-distributed tail-state episodes. On a decade of SPY data, HMM-WJ had a 97.6% in-sample KS pass rate and reduced ACF-MAE from 0.059 to 0.052 relative to the no-jump variant, although GARCH reached 0.031. This tradeoff motivated our use of the hidden Markov family as the per-asset marginal rather than establishing HMM-WJ as uniformly superior. For multi-asset composition, we showed that using a full per-asset generator draw as the residual in a single-index model adds the market variance twice, and we derived a closed-form scale correction that preserved target variance and recovered the calibrated factor loading across a 424-asset universe. Independent residuals nevertheless exaggerated daily cross-sectional dispersion and could overstate diversification gains in daily-rebalanced portfolios; setting a location-shift adjustment to a long-run prior reduced this return-level bias but did not restore the missing cross-asset dependence. By clarifying how marginal fidelity, variance preservation, and dependence interact, this framework can support the generation of more reliable multi-asset synthetic financial data for large-scale stress testing and portfolio evaluation.

2 Related Work

Models of equity-market stylized facts follow either a bottom-up or top-down approach. Bottom-up agent-based models [17–22] derive return dynamics from simulated interactions among heterogeneous agents. They are demanding to calibrate and expensive to simulate at scale. Top-down models instead fit statistical structures directly to observed growth-rate series. Top-down generators trace to Bachelier's diffusion model, later formalized as geometric Brownian motion and used by Black-Scholes-Merton for option pricing [23–26]; these diffusion baselines are tractable but fail on the empirical regularities that drive risk and pricing. Mandelbrot [3] showed returns have heavier tails than the Gaussian admits; Fama [27] reviewed evidence that raw returns have near-zero autocorrelation; and Schwert [28] documented persistence in monthly absolute residuals, later codified by Cont [4] as volatility clustering. The autoregressive conditional heteroskedasticity (ARCH) and generalized ARCH (GARCH) models of Engle and Bollerslev [6, 7] reproduced clustering but required a heavy-tailed noise term to produce heavy-tailed marginals. Neither model includes discrete regimes or sudden jumps. Stochastic-volatility models [29, 8] make volatility a latent process to capture clustering, but estimating that process requires filtering or Markov chain Monte Carlo (MCMC) methods [30]; jump-diffusion models [9, 10] inject Poisson events into the price path, producing heavy tails by construction, but volatility does not persist after the jump.

Hidden Markov models embed regime and jump behavior directly in a latent discrete chain, following Hamilton's [31] two-state regime-switching formulation, later applied to volatility forecasting [32] and equity trading [33]. Ryden et al. [15] found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but did not reproduce volatility clustering; Bulla and Bulla [16] modeled clustering with hidden semi-Markov models (HSMMs) that replaced the geometric dwell-time distribution of every state with a parametric (negative-binomial) alternative. Both rely on the Baum-Welch expectation-maximization (EM) algorithm [34], which is iterative and sensitive to initialization. Deep generative alternatives learn the marginal directly but routinely miss the volatility-clustering signature: Takahashi et al. [11] found that generative adversarial network (GAN) architectures failed to reproduce the absolute-return autocorrelation function, and Kwon and Lee [13] reported the same failure for TimeGAN [12]. Recent surveys [1, 2, 5] agree that stylized-facts reproduction remains the primary quality criterion. Our marginal component also used a latent-state chain, but it differed from the HSMM. It kept the empirically counted transition matrix for the central states and added an event-driven override that forces the chain into the tail states for a Poisson-distributed duration on a subset of time steps.

Multi-asset composition adds cross-asset dependence on top of a univariate generator. Mixture hidden Markov models [35] share latent state across assets but are costly to estimate in high dimensions. Copula approaches [36–38] separate marginals from cross-asset dependence but require choosing a copula family and, in their standard static form, treat successive time slices as independent. Factor models instead reduce a firm's growth rate to one or more shared factors [39, 40]. The single-index model (SIM) of Sharpe [41] uses one shared factor and decomposes each asset's growth rate into a market term and an idiosyncratic residual. This structure can use a single-factor market generator directly. We derived a closed-form correction that preserves the generator variance while recovering the calibrated market loading. To evaluate the resulting paths, we combined goodness-of-fit tests for the return distribution [42–44] with autocorrelation diagnostics for temporal structure [4]. These tests are necessary but not sufficient: a synthetic dataset can pass them yet still mislead when fed into the task it was generated for. Booth and Fama [45] described one such failure: a daily-rebalanced allocator on paths whose residuals are drawn independently across assets extracts a structural *diversification return* that is muted in real markets, where common sector and style factors correlate residuals across assets. We adopted the classical distributional metrics together with a Value-at-Risk coverage check scored by Kupiec's unconditional-coverage test, and addressed this diversification-return bias with a location-shift correction.

3 Methods

We fitted a marginal generator for each asset and then combined the generated paths with a variance-corrected single-index model. The dataset contained 424 United States-listed equities and exchange-traded funds observed from 2014 through 2024. The single-asset analysis focused on the SPDR S&P 500 ETF (SPY). We used 2,766 observations from January 3, 2014, through December 31, 2024, for fitting and the next 249 trading days of 2025 for out-of-sample testing. The data included daily open, high, low, and close prices and volume-weighted average prices. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from the equity price series $P_{i,j}$ as

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \tag{1}$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate derived from Treasury STRIPS (Separate Trading of Registered Interest and Principal of Securities) bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.1 Per-asset hidden Markov marginal generator

We defined the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ) as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [14] (Figure 2). The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represents a market regime, and the observation at step t is the excess growth rate G_t of Eq. (1), drawn from the continuous observation space $\mathcal{O} \subset \mathbb{R}$ (we suppress the ticker index i when only one asset is in play). The transition matrix $\mathbf{T}$ governs regime-to-regime transitions, $\mathbf{E}$ is the state-conditional emission model, and $\bar{\pi}$ is the stationary distribution. We defined the states using quantile boundaries from a fitted Laplace cumulative distribution: $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where $F_L^{-1}(\cdot;\, \mu_L, b_L)$ is the inverse cumulative distribution function of the Laplace distribution with location μ_L and scale b_L. We estimated both parameters from the in-sample growth-rate history by maximum likelihood and fixed the endpoints at $Q_0 = -\infty$ and $Q_N = +\infty$. We assigned an observation to state k when $Q_{k-1} < G_t \leq Q_k$. We chose the Laplace distribution because its sharp central peak follows the concentration of small price movements in the empirical marginal density (Fig. 1(a)) [46, 47], while its closed-form quantile function avoids iterative optimization across the 424-ticker universe [48].

The within-state emission followed a location-scale Student-t distribution:

$$G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom. For observations assigned to state k, μ_k is the sample mean and σ_k is the sample standard deviation. A standard t_ν has variance $\nu/(\nu-2)$, so σ_k is a scale parameter rather than the emission standard deviation. At $\nu = 5$, that standard deviation is $\sigma_k\sqrt{5/3}$. We selected $\nu = 5$ from a sensitivity sweep ($\nu = \infty$ recovers the Normal baseline; explored range in Table 1). At $\nu = 5$, simulated kurtosis was 7.6 against the observed 7.7, compared with 5.5 under Normal emissions; the corresponding KS pass rates were 97.6% and 97.0%. The full comparison against a Gaussian-emission HSMM baseline of Bulla and Bulla [16] is given in Table S3 of Sec. S5.

With the state space and emission family fixed, we estimated the transition matrix $\mathbf{T}$ by counting observed transitions. This avoided the initialization sensitivity of iterative expectation-maximization [48]. Direct counting assigns low probabilities to moves from central states into tail states, so the resulting model loses volatility clustering too quickly [16]. We added a jump mechanism to extend tail-state visits. Its two hyperparameters were the jump probability ϵ and the mean

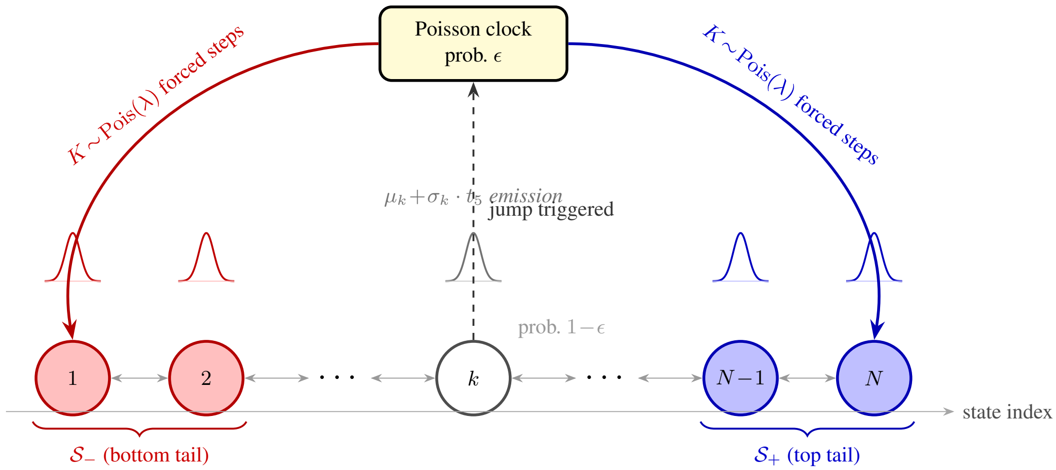

Figure 2: **Architecture of the hybrid hidden Markov model with a jump-duration mechanism (HMM-WJ).** At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1-\epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode each of the $K \sim \text{Poisson}(\lambda)$ forced steps independently draws the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) with probability p_{neg} or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) otherwise, before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

episode length λ. Tail states were defined as $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$, where N_{tail} was user-configurable (value in Table 1). When a jump occurred, we sampled its length as $K \sim \text{Poisson}(\lambda)$ [49]. Because the Poisson distribution includes zero, $K = 0$ produced no forced tail steps and the chain made an ordinary transition. Thus, λ set the mean episode length, not a minimum dwell time. During the K forced steps, the jump mechanism replaced the standard transitions with draws from a tail set. It selected $\mathcal{S}_-$ with probability p_{neg} and $\mathcal{S}_+$ otherwise (values in Table 1). The negative-tail bias reproduced the gain/loss asymmetry shown in Figure 2. The empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence, with natural residence times of 1–2 steps in every state including the tails (Fig. S5 in Online Appendix S4). The stationary distribution $\bar{\pi}$ satisfies $\bar{\pi} = \bar{\pi}\mathbf{T}$ and was estimated by matrix powering ($\mathbf{T}^{50}$) [50, 51]. We drew the initial state from $S_0 \sim \bar{\pi}$. Because $\bar{\pi}$ is the invariant distribution of $\mathbf{T}$ alone, it only approximates the invariant distribution of the jump-augmented process, whose state also includes jump status and remaining episode length; this introduces a small, unquantified deviation from exact stationarity. Algorithm 1 gives the full simulator. At each step, the chain either made a standard transition with probability $1-\epsilon$ or entered a Poisson jump episode. A jump forced K draws from a tail set before standard transitions resumed. The model sampled $G_t \sim \mathbf{E}_{S_t}$ after each transition.

We calibrated the jump probability ϵ and mean episode length λ. The mechanism followed the jump-event tradition of [9, 10], but replaced instantaneous jumps with an HSMM-style duration [16]. HMM-WJ is a discrete-time state model: the override extends visits to tail states but adds no continuous-time diffusion component. We selected (ϵ, λ) by grid search against two features of the SPY history in Fig. 1: volatility clustering and kurtosis [3, 4, 15]. The first term of the objective measured the squared error between the observed and simulated ACF of absolute excess growth rates through lag L. The second penalized the difference in global kurtosis. Most observed ACF variation occurred within the calibration window listed in Table 1; the final comparison in Fig. 3 extends to lag 252:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\text{ACF}_{\text{obs}}(\tau) - \overline{\text{ACF}}_{\text{sim}}(\tau)\right)^2 + w_\kappa \left(\kappa_{\text{obs}} - \overline{\kappa}_{\text{sim}}\right)^2, \tag{3}$$

where $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and the global kurtosis, respectively (kurtosis denoted κ to avoid collision with the jump-duration variable K), and the simulated counterparts $\overline{\text{ACF}}_{\text{sim}}(\tau)$ and $\overline{\kappa}_{\text{sim}}$ were averages over jump-active paths. We simulated

Algorithm 1 HMM-WJ single asset marginal generator.

Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon T; tail sets $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$.
Ensure: State sequence $\{S_t\}_{t=0}^{T}$ and growth-rate path $\{G_t\}_{t=1}^{T}$.

Draw $S_0 \sim \bar{\pi}$; $\quad k \leftarrow 0$ $\quad$ {k: remaining forced-step counter}
for $t = 1, \ldots, T$ **do**
  if $k > 0$ **then**
    $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $\quad S_t \sim \text{Uniform}(\mathcal{J})$; $\quad k \leftarrow k - 1$ $\quad$ {forced tail step; sign drawn per step}
  else if $\text{Uniform}(0, 1) > \epsilon$ **then**
    $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ $\quad$ {standard Markov transition}
  else
    $K \sim \text{Poisson}(\lambda)$ $\quad$ {trigger jump; K is the episode length}
    if $K = 0$ **then**
      $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$ $\quad$ {$K = 0$: no forced steps, standard transition}
    else
      $\mathcal{J} \leftarrow \mathcal{S}_-$ w.p. p_{neg}, else $\mathcal{S}_+$; $\quad S_t \sim \text{Uniform}(\mathcal{J})$; $\quad k \leftarrow \min(K - 1, T - t)$ $\quad$ {first forced step; K steps total, sign per step}
    end if
  end if
  Draw $G_t \sim \mu_{S_t} + \sigma_{S_t}\, t_5$ $\quad$ {Eq. (2)}
end for

200 independent paths of 2,766 trading days at each grid point. A path with no jump reduces to HMM-NJ and contains no information about (ϵ, λ), so the objective included only paths with at least one jump [49]. The weight w_κ balanced the kurtosis and ACF terms. The search ranges, w_κ, L, and the remaining HMM-WJ and SIM hyperparameters are reported in Table 1. The fitted model was evaluated against the marginal and temporal statistics reported in Section 4.

3.2 Variance-corrected multi-asset composition

The single-asset model describes each ticker separately. It does not say how several tickers should move together. For the multi-asset extension, we added one common source of movement: a market path. We used the single-index model (SIM) of Sharpe [41]:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{4}$$

Here g_m is the market growth-rate path, β_i measures how strongly asset i moves with the market, and ε_i is the part specific to that asset. The symbol ε_i denotes a residual path; it is unrelated to the jump probability ϵ used by the single-asset model. We estimated (α_i, β_i) from real data by ordinary least squares (OLS).

The main difficulty is choosing ε_i. A standard SIM uses Gaussian residuals with the variance implied by the real-data regression. This preserves the fitted market relationship, but it removes much of the tail and regime behavior produced by the single-asset generator. We called this baseline *Gaussian SIM*.

Why direct construction fails. A tempting alternative is to use a full synthetic return path $\tilde{g}_i$ from the single-asset generator as the residual:

$$g_i = \alpha_i + \beta_i\, g_m + \tilde{g}_i \tag{5}$$

where $\sigma^2_{\text{gen},i} = \text{Var}(\tilde{g}_i)$. We called this the *naive* construction. Its problem is double counting. The draw $\tilde{g}_i$ already has the total variance of asset i. Adding the market term contributes more variance. Because $\tilde{g}_i$ is sampled independently of g_m,

$$\text{Var}(g_i) = \beta_i^2 \sigma_m^2 + \sigma^2_{\text{gen},i}, \qquad \text{Cov}(\tilde{g}_i, g_m) \approx 0. \tag{6}$$

The excess is $\beta_i^2 \sigma_m^2$, so the error is largest for assets with high market exposure.

Variance-preserving correction. We treated $\sigma^2_{\text{gen},i}$ as a variance budget for the composed asset. The market term uses part of that budget. We scaled the single-asset draw so that it uses only the part that remains. Set $\varepsilon_i = s_i \tilde{g}_i$ and require $\mathrm{Var}(\beta_i g_m + s_i \tilde{g}_i) = \sigma^2_{\text{gen},i}$. Solving for s_i gives

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\text{gen},i}}} \tag{7}$$

For clarity, define the *market loading ratio*

$$\rho_i \;\equiv\; \frac{\beta_i^2\,\sigma_m^2}{\sigma^2_{\text{gen},i}}, \tag{8}$$

which is the fraction of the variance budget used by the market. The residual receives the remaining fraction, so $s_i^2 = 1 - \rho_i$.

This correction is impossible when $\rho_i \geq 1$ because the market term alone uses the entire variance budget. This case can occur when β_i is large, the per-asset generator has low variance, or the synthetic market is more volatile than the calibration market. In that case we reserved a minimum fraction f for the asset-specific draw and reduced the effective beta. The rule is

$$\beta_i^{\text{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \operatorname{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma^2_{\text{gen},i}}{\sigma^2_m}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{9}$$

The sign of β_i is retained. We used $f = 0.10$, which guarantees that at least 10% of the variance budget remains asset-specific (Table 1).

High-R^2 index trackers. The variance budget is the main target for ordinary stocks. For an index-tracking ETF, the fitted relationship with the market is more important. For example, SPY regressed on itself has $\beta = 1$, $R^2 = 1$, and no residual variance. QQQ and SPYG also have high R^2 against SPY. Applying the ordinary-stock rule to these trackers can preserve total variance while changing the fitted R^2.

We therefore used a separate branch when the real-data value $R^2_{i,\text{real}}$ is at least R^2_{preserve}. In our universe, 0.80 separated the index trackers from individual stocks (Fig. S3). On this branch, the residual variance required to recover the calibrated R^2 is

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1-R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{10}$$

We rescaled the generator draw to this target:

$$\varepsilon_i \;=\; \left(\sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\right)\tilde{g}_i, \tag{11}$$

When $R^2_{i,\text{real}} = 1$, the target residual variance is zero, as it should be for SPY against itself. If the scale factor in Eq. (11) exceeds one, the procedure stretches the generator draw. We flagged this case because it can weaken the original marginal fit.

Complete composition rule. Algorithm 2 puts these steps together. For a high-R^2 tracker, it preserves the calibrated R^2. For other assets, it preserves the generator variance and clips beta only when the market term would leave too little asset-specific variance. It then adds the scaled asset draw to the market term. An optional rank reorder can add dependence among the residual draws without changing their individual variances. For long paths, OLS on the composed series recovers α_i and the effective beta. The composed variance matches the target selected by the branch. These are variance and factor-loading guarantees; they do not guarantee that the complete marginal distribution is unchanged. The algebra is given in Online Appendix S8. The construction adds only same-day market dependence. It does not reproduce lead–lag effects or market-driven volatility clustering.

Algorithm 2 Hybrid SIM composition with variance correction.

Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

for each asset $i = 1, \ldots, N$ **do**
  Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \text{Var}(\tilde{g}_i)$.
  if $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
    $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}})/R^2_{i,\text{real}}$ {Eq. (10)}
    $\varepsilon_i \leftarrow \left(\sqrt{\sigma^2_{\varepsilon,\text{target}}/\sigma^2_{\text{gen},i}}\right) \tilde{g}_i$
    $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ R^2-preserving
  else
    $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ {Eq. (8)}
    if $\rho_i \leq 1 - f$ **then**
      $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; branch $\leftarrow$ variance-preserving
    else
      $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i)\sqrt{(1-f)\,\sigma^2_{\text{gen},i}/\sigma_m^2}$; branch $\leftarrow$ clipped variance-preserving
    end if
    $\varepsilon_i \leftarrow s_i \, \tilde{g}_i$
  end if
  (Optional) apply cross-sectional rank reorder to ε_i.
  $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}} g_m + \varepsilon_i$ {Eq. (4)}
end for

Table 1: **HMM-WJ and SIM composition hyperparameters.** Values used for the main SPY analysis, with the sensitivity-sweep or grid-search span and the location of the corresponding check where one exists. The 423-ticker SIM composition universe holds $\epsilon = 0$ instead of the SPY value; per-ticker (ϵ^*, λ^*) for the NVDA, JNJ, and JPM cross-asset validation are given in Table S7.

Symbol	Description	SPY value	Explored range
State partition and emission			
N	Laplace-quantile states	100	$\{30, \ldots, 350\}$, Online Appendix S6
ν	Student-t emission degrees of freedom	5	$\{3, \ldots, 30, \infty\}$
Jump-duration override			
ϵ	jump probability per step	10^{-4}	$[10^{-4}, 2.5 \times 10^{-2}]$, Eq. (3)
λ	mean jump-episode duration (steps)	100	$[10, 160]$, Eq. (3)
p_{neg}	probability a jump targets $\mathcal{S}_-$	0.52	fixed
N_{tail}	quantile states per tail set	5	fixed (at $N = 100$)
w_κ	kurtosis-penalty weight, Eq. (3)	0.20	fixed
L	ACF lag window, Eq. (3)	25 days	fixed
SIM composition			
f	idiosyncratic variance floor	0.10	$\{0.05, \ldots, 0.30\}$, Fig. S7
R^2_{preserve}	branch-selection threshold	0.80	$\{0.70, \ldots, 0.90\}$, Fig. S7

Evaluation protocol. We evaluated the method on 424 US equities and exchange-traded funds from Polygon.io. Each ticker had a complete daily-close history from January 2014 through December 2024 ($T = 2{,}767$). We used SPY as the market and treated the other 423 tickers as assets to compose. For each asset, we fitted the same state and emission model used in the SPY analysis (Table 1; $r_f = 0$). To focus the experiment on multi-asset composition rather than per-ticker jump calibration, we set the jump probability to zero. We also regressed each real asset series on SPY to estimate α_i, β_i, $R^2_{i,\text{real}}$, and residual variance. Across the universe, β ranged from 0.01 to 1.79 and $R^2_{i,\text{real}}$ ranged from 0.001 to 0.933 (Fig. S3).

We compared six ways to construct the asset-specific term. The naive method used the full JumpHMM draw without correction. The Gaussian SIM method used independent Gaussian residuals with variance estimated by OLS. The hybrid method used Algorithm 2. The JumpHMM-on-residuals method fitted a new JumpHMM to the OLS residual series after converting the residuals to a synthetic price path by cumulative compounding. The block-bootstrap method sampled the OLS residual series with the Politis–Romano stationary bootstrap [52] and a mean block length of $\sqrt{T} \approx 50$. The GARCH(1,1)-t method fitted a conditional-variance model to each residual series [7]. We excluded thirty non-stationary fits from the GARCH results only. For each ticker, the naive and hybrid methods received the same JumpHMM draw. This pairing isolated the effect of the variance correction. The other four methods generated their own residual draws. We did not use the optional cross-sectional rank reorder because the experiment tested each ticker's marginal construction separately. We evaluated whether OLS recovered the calibrated β and R^2, whether the synthetic series matched the real return distribution, and whether the composition preserved the generator variance. We measured these outcomes with $|\hat{\beta} - \beta|$, $|\hat{R}^2 - R^2_{\text{real}}|$, KS and AD tests, Wasserstein-1 distance, excess kurtosis, the Hill upper-tail index, and the ratio of composed to generator variance. Definitions are in Online Appendix S3. We ran 100 replications per ticker with seed 1234, producing 42,300 series per method. Pass rates used $\alpha = 0.05$.

4 Results

4.1 Marginal-generator validation

We first measured the empirical SPY data targets that any synthetic generator must reproduce. In both the in-sample (2014–2024, $T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows, the real SPY series had excess kurtosis with Jarque-Bera rejection, persistent volatility clustering, and near-zero linear autocorrelation in raw returns (Table S1) [27]. We calibrated (ϵ, λ) for HMM-WJ against the in-sample SPY series using the grid search in Eq. (3) (Algorithm 1; search range in Table 1). The selected values were $(\epsilon, \lambda) = (10^{-4}, 100)$, although nearby grid points gave similar objective values. At this setting, HMM-WJ had a 97.6% KS pass rate, a 91.3% AD pass rate, and an ACF-MAE of 0.052 (Table 2). The primary analysis used $N = 100$ states. Across $N = 60$ to 200, the KS and AD pass rates varied by less than one percentage point for both HMM-NJ and HMM-WJ. At $N = 350$, some states were never visited and their emission parameters could not be estimated (Online Appendix S6).

With the model calibrated, we benchmarked HMM-WJ (Algorithm 1) against seven alternative generators on 1,000 synthetic paths of 2,766 trading days each (Table 2, Figure 3). Bootstrap resampling had 100% KS and 100% coverage, while Gaussian sampling had 0% KS. GARCH(1,1) had the lowest ACF-MAE (0.031) and a 5.5% KS pass rate. The Gated Recurrent Unit (GRU) [53] achieved an ACF-MAE of 0.036 but a 0.6% KS pass rate. The hidden semi-Markov model (HSMM) [16] reached 82% KS. Its 8-state partition produced simulated kurtosis of 4.8 against the observed 7.7, with 68.7% quantile coverage. A comparison of persistence specifications (HMM-WJ vs HSMM) and emission families (Gaussian vs Student-t) is given in Table S3 (Sec. S5). HMM-NJ and HMM-WJ had KS pass rates of 99.7% and 97.6%, respectively, and AD pass rates of 99.1% and 91.3%. HMM-WJ's pass rates ran 2–8 percentage points below HMM-NJ's because jumps occasionally overweight tail states relative to their empirical frequency. The Student-t ($\nu = 5$) emission produced simulated kurtosis of 7.6 against the observed 7.7, while Normal emissions underestimated it by $\sim 29\%$ (Table S3). Both HMM variants reproduced the observed negative skewness of -0.75 (Table S1). This skewness came from asymmetric occupancy of the Laplace states, not an explicit skewness parameter. The symmetric-innovation models (GARCH, Gaussian, and Laplace) produced near-zero skewness by construction. For temporal dependence, HMM-WJ reduced ACF-MAE from 0.059 for HMM-NJ to 0.052. GARCH still had the lowest ACF-MAE at 0.031, and only HMM-WJ, among the non-GARCH, non-neural models, scored below the shared i.i.d. value of 0.060, driven by the $\sim 24\%$ of paths that contained at least one Poisson jump and sustained $\mathrm{ACF}(|G_t|)$ decay across all lags. The jump override was inactive in the remaining 76%, for which HMM-WJ reduces to HMM-NJ. On the held-out 249 days of 2025, HMM-NJ and HMM-WJ achieved 96.7% and 94.4% KS with simulated kurtosis of 6.4 and 6.1 against an observed 6.9, while GARCH's W_1 rose by 67% from its in-sample value (Table 2; Figure S4). The tail Q–Q plot for all eight generators is shown in Fig. 3c. Cross-asset evaluation on NVDA, JNJ, and JPM produced HMM-WJ in-sample KS pass rates of at least 98.9%

Table 2: **In-sample and out-of-sample model comparison for SPY** (1,000 simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from distributions fitted by maximum likelihood estimation (MLE). GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. GRU: 2-layer gated recurrent unit [53] with Gaussian output head. HSMM: hidden semi-Markov model with $K = 8$ Laplace quantile states, negative-binomial dwell times, and Student-t ($\nu = 5$) emissions [16]. HMM-NJ: hidden Markov model ($N = 100$ states) without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Wasserstein-1 and Hellinger distance defined in Online Appendix S3. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors (SE) in parentheses: binomial SE for pass rates, $\mathrm{std}/\sqrt{n}$ for kurtosis, Wasserstein-1, and Hellinger; bootstrap SE ($B = 500$) for ACF-MAE and coverage. Best value per row in bold.

Metric	**Bootstrap**	**Gaussian**	**Laplace**	**GARCH(1,1)**	**GRU**	**HSMM**	**HMM-NJ**	**HMM-WJ**
In-sample: 2,766 *trading days* (2014–2024)								
KS pass rate (%) ↑	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	0.6 (0.2)	82.0 (1.2)	**99.7** (0.2)	97.6 (0.5)
AD pass rate (%) ↑	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	0.2 (0.1)	42.5 (1.6)	**99.1** (0.3)	91.3 (0.9)
Excess kurtosis (observed)				7.715				
Excess kurtosis (simulated) ≈	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	**8.2** (0.37)	5.7 (0.16)	4.8 (0.08)	8.1 (0.14)	7.6 (0.14)
ACF-MAE ↓	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	**0.031** (0.001)	0.036 (<0.001)	0.059 (<0.001)	0.059 (<0.001)	0.052 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.4)	17.2 (0.5)	68.7 (1.1)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.062 (0.001)	0.399 (0.001)	0.138 (0.001)	0.304 (0.006)	0.421 (0.003)	0.176 (0.001)	**0.081** (0.001)	0.101 (0.001)
Hellinger dist ↓	0.042 (<0.001)	0.148 (<0.001)	**0.072** (<0.001)	0.100 (0.001)	0.134 (0.001)	0.113 (<0.001)	0.073 (<0.001)	0.075 (<0.001)
Out-of-sample: 249 *trading days* (2025)								
KS pass rate (%) ↑	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	71.8 (1.4)	96.2 (0.6)	**96.7** (0.6)	94.4 (0.7)
AD pass rate (%) ↑	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	44.8 (1.6)	**96.7** (0.6)	**96.7** (0.6)	95.1 (0.7)
Excess kurtosis (observed)				6.867				
Excess kurtosis (simulated) ≈	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	2.9 (0.10)	4.1 (0.13)	**6.4** (0.22)	6.1 (0.13)
ACF-MAE ↓	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	**0.026** (<0.001)	0.033 (<0.001)	0.042 (<0.001)	0.041 (<0.001)	0.039 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	36.4 (1.9)	74.7 (1.0)	96.0 (1.3)	77.8 (2.9)	**100.0** (0.2)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.232 (0.002)	0.452 (0.002)	0.263 (0.002)	0.507 (0.018)	0.488 (0.005)	0.287 (0.002)	**0.258** (0.002)	0.282 (0.006)
Hellinger dist ↓	0.205 (0.001)	0.235 (0.001)	0.211 (0.001)	0.232 (0.001)	0.240 (0.001)	0.239 (0.001)	**0.207** (0.001)	0.210 (0.001)

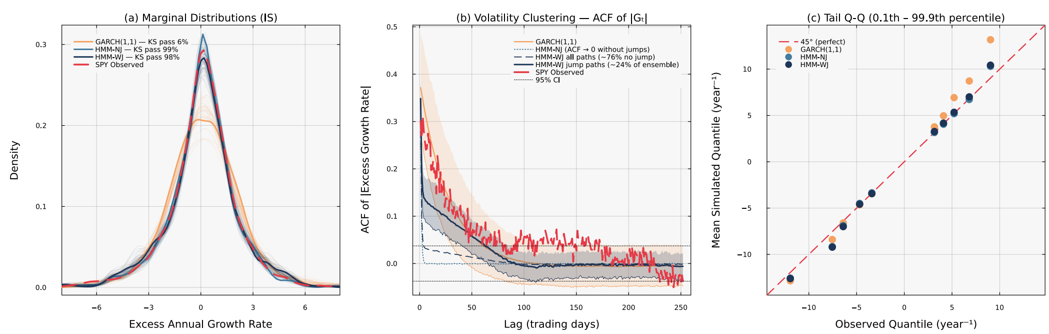

Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample KS pass rates annotated per generator. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths) per generator. *Panel (c):* tail Q-Q plot at the 0.1st–99.9th percentile region per generator.

on all three. The out-of-sample pass rates were 97.7% for NVDA, 74.0% for JNJ, and 79.2% for JPM (Table S7, Online Appendix S7).

To measure how jump-active paths changed the results, we resampled the fitted ensemble at different jump-path fractions f (Figure 4). Here f is the share of accepted paths that contain at least one jump. We varied f from 0, where all accepted paths reduce to HMM-NJ, to 1, where every accepted path contains a jump. This procedure held the fitted model fixed and changed only the mixture of accepted paths. The absolute-return autocorrelation error fell monotonically from 0.059 at $f = 0$ to 0.037 at $f = 1$, moving from the no-jump value toward the GARCH value as the jump share rose. Over the same sweep, simulated excess kurtosis declined from 7.9 to 6.7. It matched the observed value near $f \approx 0.14$ and fell below it thereafter. The two-sample KS and AD pass rates also declined. The Hill upper-tail index stayed near 0.35, against an observed 0.32, across the whole sweep. The Student-t ($\nu = 5$) emission sets the tail exponent independently of jump content. Increasing the jump share therefore traded fit to the distributional body for lower autocorrelation error without changing the tail

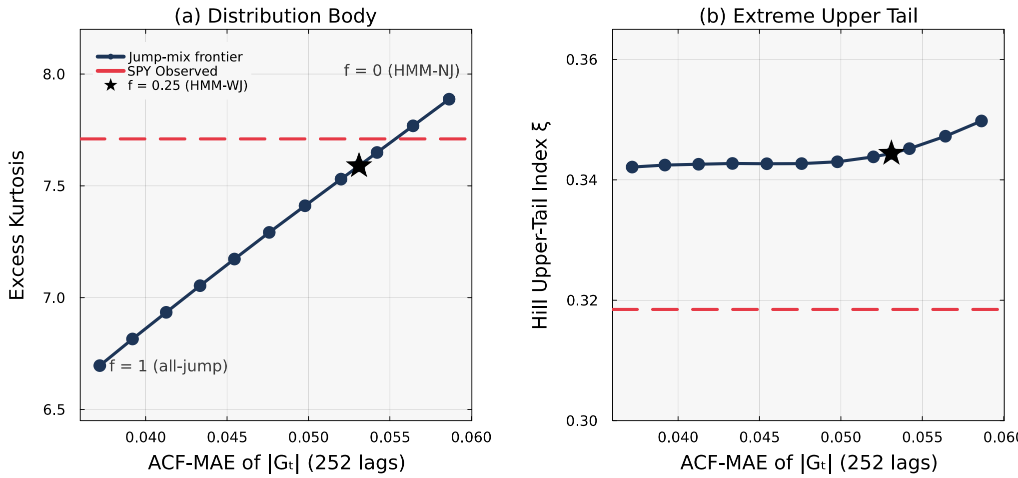

Figure 4: **Jump-mix enrichment frontier for SPY** (in-sample, $N = 100$, pool of $4{,}000$ simulated paths). Resampling the fitted HMM-WJ ensemble to a target jump-path fraction f (the share of accepted paths containing at least one jump) traces the temporal-marginal tradeoff. Both panels share the horizontal axis: absolute-return autocorrelation error (ACF-MAE of $|G_t|$ over lags 1–252); lower values indicate stronger volatility clustering. Panel (a): simulated excess kurtosis. Panel (b): Hill upper-tail index ξ, the raw estimator equal to the reciprocal of the tail exponent. Red dashed lines mark the observed SPY values; the star marks the $f \approx 0.25$ value used throughout. Endpoints $f = 0$ and $f = 1$ correspond to the no-jump generator and an all-jump ensemble.

Table 3: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composition method. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

index. The value $f \approx 0.25$ used throughout produced simulated kurtosis of 7.59 against the observed 7.72 and ACF-MAE of 0.053 against the no-jump value of 0.059.

4.2 Variance-corrected multi-asset composition

The single-asset results defined the tradeoff in the marginal generator. We then tested whether the variance-corrected SIM preserved each asset's variance and factor loading across the 423-ticker universe. We compared the six methods in Table 3. The median absolute β error ranged from 0.018 to 0.022 across methods. The methods differed more in R^2 recovery, marginal fit, and tail behavior. Gaussian SIM had the lowest median R^2 error (0.008) because it targets R^2 by construction. Its KS pass rate was 0.6%, its AD pass rate was 0.5%, and its excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. S2). Naive composition retained the generator marginal but inflated its variance by $\beta_i^2\sigma_m^2$, so its KS and AD pass rates were 4.5% and 2.7%. The hybrid method had a 50.4% KS pass rate, a 47.6% AD pass rate, kurtosis of 7.2, and a Hill index of 0.36. Its median R^2 error was 0.021, between the errors for naive composition and Gaussian SIM, because the variance-preserving branch targets $\sigma^2_{\text{gen},i}$ rather than $R^2_{i,\text{real}}$. The three residual-fit methods sidestepped the variance-inflation problem entirely by fitting a generator directly to the empirical OLS residual series e_i. The residual-series variance $\sigma^2_{\varepsilon,\mathrm{real},i}$ excludes the market component by construction. Their aggregate KS pass rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4%. The residual-fit methods also had lower Wasserstein distances (0.232–0.267) than hybrid (0.272). Hybrid's median excess kurtosis and Hill upper-tail index were 7.17 and 0.365. The corresponding values for JumpHMM-on-residuals were 7.26 and 0.365. Block bootstrap and GARCH, by fitting their own residual generators, produced lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based method.

4.3 VaR coverage

We tested the six methods with a per-ticker, one-day Value-at-Risk (VaR) coverage check (Table 4). At $\alpha = 0.99$, the nominal exceedance rate is 1%. The empirical rates were 0.67% for naive composition and 1.40% for Gaussian SIM. Their Kupiec pass rates were 45% and 47%, respectively. The hybrid, JumpHMM-on-residuals, block-bootstrap, and GARCH(1,1)-t methods had exceedance rates of 0.95%, 0.99%, 1.09%, and 1.09%, with Kupiec pass rates between 72% and 86%. Hybrid and JumpHMM-on-residuals differed by 0.04 percentage points, with cross-ticker SDs of 0.27 and 0.26 percentage points.

4.4 Sensitivity analyses

The KS differences did not carry over to the VaR results. We therefore plotted each ticker's KS pass rate and variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ against its calibrated β (Fig. 5). Under naive composition, the variance ratio followed $1 + \rho_i$ and exceeded generator variance by 50–75% at high β. The hybrid method held the ratio at one. Over the same range, the naive KS pass rate fell from about 0.25 at $\beta \approx 0.3$ to below 0.01 above $\beta \approx 0.7$, while the hybrid rate remained between 0.3 and 0.9. The branch-selection rule assigned 421 tickers to the variance-preserving branch, two trackers to the R^2-preserving branch, and none to the clipped branch (Fig. S6, Table S4). QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$) were the two R^2-preserving assignments, and both had a 99% KS pass rate. The highest-R^2 individual stock was BLK at 0.645, about 0.16 below the $R^2_{\mathrm{preserve}} = 0.80$ threshold. Clipping did not occur in the universe: $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\mathrm{gen},i}$ (Eq. (8)) stayed below $1 - f = 0.90$ for every ticker. Because the R^2-preserving branch contained only two tickers, we tested it on a synthetic-tracker grid spanning $(\beta_{\mathrm{true}}, R^2_{\mathrm{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$. Every cell passed KS.

Bucketing the universe by β quartile, the hybrid KS pass rate fell from 67% in the lowest quartile to 44% in the highest (Table S5). As β increased, the market term accounted for more of the composed variance, making the marginal more dependent on the market distribution. We tested the clipping rule directly by rescaling the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals

Table 4: **Per-ticker one-day Value-at-Risk coverage check across the six composition methods.** For each (ticker, replication) pair we estimated the VaR threshold at coverage level α from the composed path and counted exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1 - \alpha$). *SD*: cross-ticker standard deviation of the exceedance rate, in percentage points. *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals differ by less than the cross-ticker spread at both coverage levels; naive composition and Gaussian SIM fail coverage on a majority of the universe.

	$\alpha = 0.95$			$\alpha = 0.99$		
Composer	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	3.55	0.52	10.4	0.67	0.23	45.2
Gaussian SIM	4.06	0.76	42.6	1.40	0.29	47.0
Hybrid	4.75	0.56	80.4	0.95	0.27	79.7
JumpHMM-on-residuals	4.87	0.55	84.0	0.99	0.26	82.9
Block bootstrap	4.78	0.64	76.7	1.09	0.25	86.0
GARCH(1,1)-t	4.64	0.93	69.0	1.09	0.35	72.1

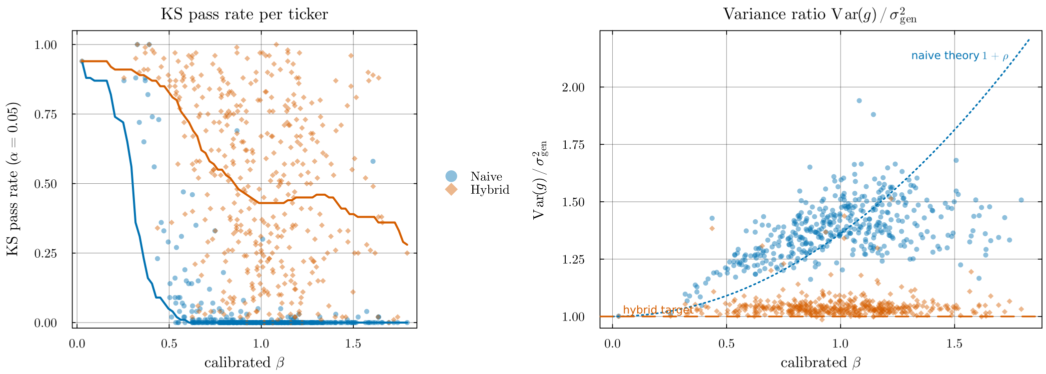

Figure 5: **Hybrid's variance correction and its KS-pass-rate consequence, per ticker vs calibrated β.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$ for naive composition (blue) and hybrid (orange). *Right (cause):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\text{gen}}$ for the same two methods. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the dashed line marks a variance ratio of one.

fixed (Table S6). At $\gamma = 1$ no ticker crossed the threshold, and the hybrid KS pass rate was 50.3%. At $\gamma = 2$ and 3, clipping applied to 76.4% and 95.2% of tickers, and the KS pass rate rose to 71.8% and 77.3%.

Finally, we tested the aggregate hybrid KS pass rate of Table 3 against the two hyperparameter choices of Algorithm 2 (Table 1), re-running hybrid on the full 423-ticker universe at every point of the $R^2_{\text{preserve}} \times f$ grid in Fig. S7. The hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 grid cells, a spread of 0.14 percentage points. QQQ and SPYG remained in the R^2-preserving branch for every threshold at or below 0.85, and the floor f did not change any branch assignment. We then measured the contribution of generator stochasticity to the aggregate metrics by re-running the full six-method protocol under 8 random seeds (Table S8). Per-method KS pass rates held to within fractions of a percentage point across seeds, and the seed sweep reproduced the relative ordering of the methods without inversions. For example, the 25 percentage-point KS difference between hybrid and JumpHMM-on-residuals exceeded hybrid's across-seed SD of 0.18 percentage points.

5 Discussion

The single-asset analysis supplied the marginal input for the multi-asset construction. Relative to HMM-NJ, HMM-WJ reduced ACF-MAE at the cost of somewhat lower in-sample KS and AD pass rates. GARCH reached the lowest ACF-MAE of any generator but failed the distributional tests, and the GRU behaved the same way. The HSMM passed the KS test but underproduced kurtosis relative to the observed data. Bootstrap resampling passed KS trivially by drawing only observed returns. Among the models that were neither GARCH nor neural, HMM-WJ was the only one with ACF-MAE below the i.i.d. baseline, and the same ordering held in the 2025 evaluation. We therefore treated HMM-WJ as a marginal generator that traded lower distributional pass rates for lower ACF-MAE, not as a uniformly better single-asset model.

The pass-rate comparisons have a statistical limitation. Two-sample KS and AD p-values assume independent observations, but the HMM, HSMM, GARCH, and bootstrap generators produce dependent observations. We therefore considered the pass rates alongside W_1, Hellinger distance, and the Hill tail index rather than comparing their p-values on the same footing. The $N = 100$ state resolution, selected by a sensitivity sweep over $\{30, 60, 90, 100, 150, 200\}$, produced little KS or AD variation across the middle of that range on the 2,766-day SPY sample. Shorter histories will support fewer states because each state will receive fewer observations. The jump parameters (ϵ, λ) were tuned once against SPY and applied across the universe. Per-ticker tuning on the three cross-asset tickers produced in-sample KS pass rates of at least 98.9% (Online Appendix S7). These three cases do not show that the SPY calibration will transfer to assets with different jump behavior. The Student-t emission with $\nu = 5$ matched the observed kurtosis far more closely than Normal emissions, but it does not model skewness directly. The negative skewness in both HMM variants came from asymmetric occupancy of the Laplace states. Assets with a different skew direction may require a different quantile rule.

The main result was the multi-asset variance correction. Naive composition inflated generator variance at high β and almost always failed the KS test, as did the Gaussian SIM. The corrected method held the variance ratio at one and passed the KS test about half the time. The three residual-fit methods had higher KS pass rates than the hybrid method. These methods fit the OLS residuals directly and therefore avoid the variance-inflation problem. The choice between the two approaches depends on whether the same full-return generator must serve both single-asset and multi-asset uses. The hybrid method's one-day 99% VaR exceedance rate sat near the 1% nominal level, comparable to the residual-fit methods, and it did not require their separate fitting stage, though its marginal KS pass rate was lower.

Several evaluation choices limit these results. We fixed the market path at the real SPY history. The naive and hybrid methods used the same JumpHMM draw for each ticker, while the other methods drew from their own generators. We also omitted the optional cross-sectional rank reorder and evaluated VaR per ticker rather than at the portfolio level. Portfolio-level VaR would require a model of cross-asset dependence. The variance correction assumes $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, which holds by construction here because the JumpHMM marginals were fitted and sampled independently of the market path. The correction would need adjustment for a generator already correlated with the market (for example, a bootstrap from joint history). The construction also injects contemporaneous market dependence only: lagged effects such as lead-lag correlation or market-driven volatility clustering are not reproduced. Clipping did not trigger because every ticker had $\rho_i < 0.90$. It becomes relevant when the simulated market is more volatile than the calibration history or for universes mixing low-volatility stocks with extreme-beta assets such as leveraged ETFs. Even without clipping, the hybrid KS pass rate declined from the lowest β quartile to the highest. As β increased, the market term accounted for more of the composed distribution, whose tails then depended more on the market path.

Independent residuals produced a diversification-return bias under daily rebalancing. Matching each asset's marginal variance did not restore the residual covariance created by common sector and style factors [45]. The resulting excess cross-sectional dispersion became a spurious return premium under rebalancing. This problem affected every method that drew residuals independently, not only naive composition. The block-bootstrap and GARCH methods restored each ticker's temporal structure but not covariance between tickers. In a 20-asset target-weight allocator example, the synthetic median annualized growth rate exceeded the real 2025 return by about 18 percentage points. The static-versus-daily-rebalance difference was 27.73 percentage points, whereas adding the allocation signal changed the median by 0.13 percentage points [45, 54, 55] (Online Appendix S2). We applied a location-shift correction that subtracts a constant drift in log-wealth from every path, calibrated to match an external long-run prior [56, 57] rather than the realized test-year outcome. The correction preserved path ranks but changed drawdown dates and depths because the adjustment varies over time in wealth space. It recalibrated the return level but did not restore cross-asset covariance. The full method, the rejected alternatives, and the percentile placement of the realized year are given in Online Appendix S2.

Several extensions remain open. Per-ticker tuning of the jump parameters (ϵ, λ) on each asset's own history would require a separate calibration for every asset. Multi-factor or lagged-factor variants could add the dependence omitted by the current single-factor construction. The same variance correction extends to either case, but we did not evaluate those extensions. A joint residual model, such as a synchronized multivariate block bootstrap, a residual factor model, or a copula, could restore cross-asset residual covariance. In contrast, separate autoregressive or block-bootstrap models would improve each asset's temporal dependence without creating dependence between assets. A stress test that pairs the clipping rule with a market more volatile than the calibration history would test whether the idiosyncratic floor keeps per-asset marginals heavy-tailed under exactly the conditions where naive SIM composition breaks down.

6 Conclusion

In this study, we developed a variance-corrected method for composing per-asset generators through a single-index model. The correction removed the variance inflation caused by inserting a full-return generator draw into the market factor. Across 423 non-market assets, it recovered the calibrated factor loading to a low median absolute error and produced heavy-tailed marginals whose per-ticker one-day 99% Value-at-Risk exceedance rate sat near the 1% nominal level, comparable to the residual-fit methods. Those residual models, fitted directly to the OLS residual series, reached higher KS pass rates than the corrected method but required a separate fit for each asset.

The per-asset component was a hidden Markov marginal model with Laplace-quantile states, Student-t emissions, and direct transition counting. On SPY, its jump-duration variant reduced absolute-return autocorrelation error relative to the no-jump variant while keeping high KS and AD pass rates; GARCH reached a lower autocorrelation error but failed the distributional tests. The single-asset result was therefore a measured marginal–temporal tradeoff rather than a claim of uniform superiority.

The current multi-asset implementation does not model cross-asset residual covariance. In a 20-asset allocator, independent residuals inflated the synthetic median daily-rebalanced return well above the realized 2025 outcome, almost entirely through the static-to-daily-rebalance diversification return rather than the allocation signal. A location-shift correction set to a documented long-run prior adjusted the return level without restoring the missing covariance. A full joint residual model is needed before using these paths for portfolio-level return forecasts.

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Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. The experiment scripts, evaluation harness (six-composer benchmark, Value-at-Risk coverage check, bias-correction utility), and the fitted per-asset marginals are released in the paper repository (`https://github.com/varnerlab/HMM-w-jumps-paper`); the underlying model-fitting library is released separately as `JumpHMM.jl` (`https://github.com/varnerlab/JumpHMM.jl`), pinned as a dependency of the experiment code.

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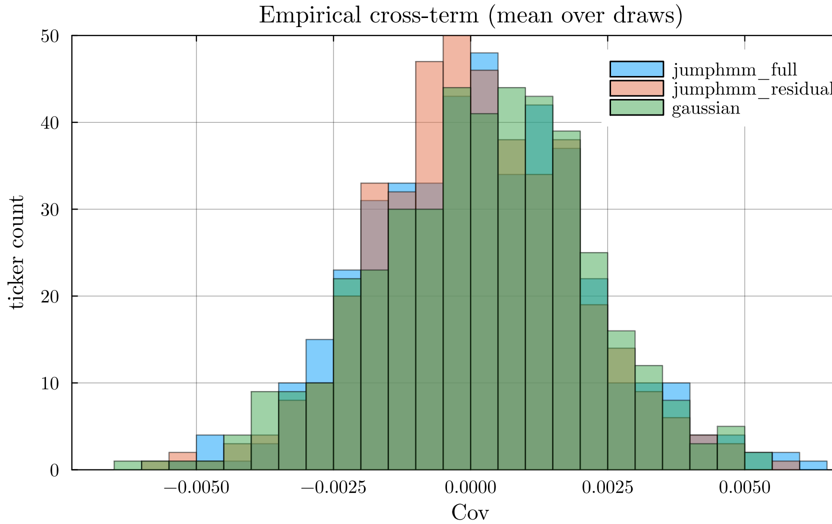

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}(\tilde{g}_i, g_m)/(\sigma_{\mathrm{gen},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. Universe means were 1.53×10^{-4}, 6.58×10^{-5}, and 2.26×10^{-4} for the full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators.

S1 Empirical cross-term diagnostic

The variance correction of Eq. (7) assumes $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, allowing the variance derivation to omit the cross-term. This assumption does not follow automatically from the fitting procedure because each JumpHMM uses the asset's full growth-rate history, including its market-correlated component. A remaining market loading in the generator draw would invalidate the closed-form scale. We tested the assumption by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R} \frac{\mathrm{Cov}(\tilde{g}_i^{(r)},\, g_m)}{\sigma_{\mathrm{gen},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values were plotted as histograms in Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composition methods), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composition method), and the Gaussian baseline had universe-mean normalized cross-covariances of 1.53×10^{-4}, 6.58×10^{-5}, and 2.26×10^{-4}, respectively. Their per-ticker values ranged from -0.0061 to 0.0061. These values quantify the cross-term omitted in Eq. (7).

S2 Bias correction for daily-rebalanced backtests

This appendix derives the bias correction in Eq. (S14). Under daily rebalancing, the raw synthetic ensemble had a median annualized growth rate about 22 percentage points above the long-run prior. The variance correction in Eq. (7) first sets each synthetic asset's second moment to its empirical target. The second-order formula for the diversification return is therefore the same for the synthetic and real marginals. The remaining 22-percentage-point difference comes from omitted cross-asset residual covariance, higher-order Taylor terms, calibration drift, and the documented attenuation of realized diversification returns [54], which the location-shift correction absorbs as a single empirical drag on log-wealth.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence but discard temporal structure in the residual draw: at each trading day the per-asset residual $\varepsilon_i(t)$ is sampled independently from the previous day. Real equity residuals have small but non-negligible serial dependence at sub-monthly horizons [4], so the i.i.d.-residual generator family used here drops a real feature of the data. Consider a daily-rebalanced portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t)\,\Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t \;=\; W_{t-1} \sum_i w_i \exp\!\big(g_i(t)\,\Delta t\big), \tag{S2}$$

so the per-step log return on the portfolio is $\log(W_t/W_{t-1}) = \log\sum_i w_i \exp(g_i(t)\,\Delta t)$. We suppress the time argument, writing g_i for $g_i(t)$. A second-order Taylor expansion of the exponential about zero, $\exp(g_i\,\Delta t) = 1 + g_i\,\Delta t + \frac{1}{2}(g_i\,\Delta t)^2 + O((g\,\Delta t)^3)$, substituted into each summand of $\sum_i w_i \exp(g_i\,\Delta t)$ gives:

$$\begin{aligned}
\sum_i w_i \exp\!\big(g_i\,\Delta t\big) &= \sum_i w_i \big[\,1 + g_i\,\Delta t + \tfrac{1}{2}(g_i\,\Delta t)^2 + O\big((g\,\Delta t)^3\big)\big] && \text{(Taylor)}\\
&= \sum_i w_i + \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 + O\big((g\,\Delta t)^3\big) && \text{(distribute)}\\
&= 1 + \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 + O\big((g\,\Delta t)^3\big). && \big(\textstyle\sum_i w_i = 1\big)
\end{aligned} \tag{S3}$$

A second-order Taylor expansion of $\log(1+x)$ about $x = 0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1 + x$ with the right-hand side of Eq. (S3), the order-Δt correction is $x := \Delta t \sum_i w_i\, g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2(\sum_i w_i\, g_i)^2 + O((\Delta t)^3)$, since the cross term has an extra Δt factor. Substituting:

$$\begin{aligned}
\log\frac{W_t}{W_{t-1}} &= \log\Big(1 + x + O\big((g\,\Delta t)^3\big)\Big) && \text{(by Eq. (S3))}\\
&= x - \tfrac{1}{2}\,x^2 + O\big(x^3\big) && \text{(Taylor)}\\
&= \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\, g_i^2 - \tfrac{1}{2}(\Delta t)^2\Big(\sum_i w_i\, g_i\Big)^2 + O\big((g\,\Delta t)^3\big) && \text{(plug in)}\\
&= \Delta t \sum_i w_i\, g_i + \tfrac{1}{2}(\Delta t)^2 \left(\sum_i w_i\, g_i^2 - \Big(\sum_i w_i\, g_i\Big)^2\right) + O\big((g\,\Delta t)^3\big). && \text{(group)}
\end{aligned} \tag{S4}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1}\log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p \;=\; \sum_i w_i\, g_i \;+\; \tfrac{1}{2}\,\Delta t \left(\sum_i w_i\, g_i^2 \;-\; \Big(\sum_i w_i\, g_i\Big)^2\right) \;+\; O\big((\Delta t)^2\big). \tag{S5}$$

Restoring the time argument, we now take unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i \;:=\; \mathbb{E}\big(g_i(t)\big), \qquad \sigma_i^2 \;:=\; \mathrm{Var}\big(g_i(t)\big), \tag{S6}$$

the per-rate covariance $\Sigma_{ij} := \mathrm{Cov}\big(g_i(t), g_j(t)\big)$, and the per-rate portfolio variance $\sigma_p^2 := \mathrm{Var}\big(\sum_i w_i g_i(t)\big) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i\, g_i(t)^2$ and $(\sum_i w_i\, g_i(t))^2$. Both expand by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}\big(X^2\big) \;=\; \mathbb{E}\big(X\big)^2 \;+\; \mathrm{Var}(X). \tag{S7}$$

Applied to $X = g_i(t)$, with $\mathbb{E}(g_i) = \mu_i$ and $\mathrm{Var}(g_i) = \sigma_i^2$, then summed against w_i by linearity of expectation:

$$\begin{aligned}
\mathbb{E}\Big(\sum_i w_i\, g_i(t)^2\Big) \;&=\; \sum_i w_i\, \mathbb{E}\big(g_i(t)^2\big) && \text{(linearity)} \\
&=\; \sum_i w_i\big[\,\mu_i^2 + \sigma_i^2\,\big] && \text{(by Eq. (S7))} \\
&=\; \sum_i w_i\, \mu_i^2 \;+\; \sum_i w_i\, \sigma_i^2. && \text{(S8)}
\end{aligned}$$

Applied to $X = \sum_i w_i g_i(t)$, with $\mathbb{E}(X) = \sum_i w_i\, \mu_i$ by linearity and $\mathrm{Var}(X) = w^\top \Sigma w = \sigma_p^2$ by definition of σ_p^2:

$$\begin{aligned}
\mathbb{E}\Big(\Big(\textstyle\sum_i w_i g_i(t)\Big)^2\Big) \;&=\; \mathbb{E}\Big(\textstyle\sum_i w_i g_i(t)\Big)^2 \;+\; \mathrm{Var}\Big(\textstyle\sum_i w_i g_i(t)\Big) && \text{(by Eq. (S7))} \\
&=\; \Big(\textstyle\sum_i w_i\, \mu_i\Big)^2 \;+\; \sigma_p^2. && \text{(S9)}
\end{aligned}$$

Substituting Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouping the σ_i^2 contributions separately from the μ_i contributions:

$$\mathbb{E}(g_p) \;=\; \sum_i w_i\, \mu_i + \underbrace{\tfrac{1}{2}\,\Delta t\Bigg(\sum_i w_i\, \sigma_i^2 - \sigma_p^2\Bigg)}_{D} + \tfrac{1}{2}\,\Delta t\Bigg(\sum_i w_i\, \mu_i^2 - \Big(\sum_i w_i\, \mu_i\Big)^2\Bigg) + O\big((\Delta t)^2\big), \tag{S10}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units [45, 55]. The middle term, D, is the diversification return: the $O(\Delta t)$ correction equal to half the difference between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2. The per-rate moments μ_i and σ_i^2 have units yr^{-1} and yr^{-2} respectively; the products $\Delta t\, \sigma_i^2$ and $\Delta t\, \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}). The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t\, \mu^2 \sim 3\times10^{-5}\ \mathrm{yr}^{-1}$ at $\mu \sim 0.08\ \mathrm{yr}^{-1}$ and negligible against the variance term $D \sim \Delta t\, \sigma^2 \sim 4\times10^{-2}\ \mathrm{yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T\,\Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + 2\sum_{i<j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i(1 - w_i)\sigma_i^2 = \sum_{i<j} w_i w_j(\sigma_i^2 + \sigma_j^2)$ that holds for any weights summing to one:

$$D \;=\; \tfrac{1}{2}\,\Delta t\Bigg(\sum_i w_i \sigma_i^2 - \sigma_p^2\Bigg) \;=\; \tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j\ \mathrm{Var}(g_i - g_j) \;\geq\; 0, \tag{S11}$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i, j) comoves perfectly ($\mathrm{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The rebalancing rule itself produces this effect: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz [55] described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama [45] called it the diversification return.

Substituting the single-index decomposition $g_i(t) = \alpha_i + \beta_i\, g_m(t) + \varepsilon_i(t)$, with market variance $\sigma_m^2 := \mathrm{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \mathrm{Var}(\varepsilon_i(t))$, and $\mathrm{Cov}(g_m, \varepsilon_i) = 0$ (verified in Sec. S1), into Eq. (S11) gives:

$$\mathrm{Var}(g_i - g_j) \;=\; (\beta_i - \beta_j)^2 \sigma_m^2 \;+\; \mathrm{Var}(\varepsilon_i - \varepsilon_j), \tag{S12}$$

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample (IS): 2014–2024 ($T = 2{,}766$); out-of-sample (OoS): 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. JB: Jarque-Bera normality test. LB: Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed	Gaussian	Laplace
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	-0.753	-1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $\lvert G_t \rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

since g_m and the residuals are uncorrelated, so the diversification return splits into a market-beta-dispersion component and an idiosyncratic component:

$$D \;=\; \underbrace{\tfrac{1}{2}\,\Delta t\,\sigma_m^2 \sum_{i<j} w_i w_j (\beta_i - \beta_j)^2}_{D_{\mathrm{mkt}}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j \;\mathrm{Var}(\varepsilon_i - \varepsilon_j)}_{D_\varepsilon}, \tag{S13}$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2}\,\Delta t\,\sigma_m^2[\sum_i w_i\beta_i^2 - (\sum_i w_i\beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t\,\sigma_m^2$. Across the 423-ticker universe the empirical β distribution spans $[0.01, 1.79]$ with interquartile range $[0.81, 1.18]$ and the SPY annualized variance $\Delta t\,\sigma_m^2 \approx 0.04\ \mathrm{yr}^{-1}$. The weighted variance of β is bounded by one quarter of its squared range, so D_{mkt} is at most $0.016\ \mathrm{yr}^{-1}$, or 1.6 percentage points per year over this universe. For the idiosyncratic component D_ε, residuals were approximately orthogonal across assets at the same time slice. Thus, $\mathrm{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_ε scales with the weighted-average annualized idiosyncratic variance $\Delta t\,\sigma_{\varepsilon,i}^2$ per name.

The second-order value of D depends on same-time unconditional moments, not on serial dependence. Equation (7) sets $\sigma_{\mathrm{gen},i}^2 = \beta_i^2\sigma_m^2 + \sigma_{\varepsilon,i}^2$ to its empirical target. Any residual generator with the same unconditional second moments therefore gives the same D. This includes a stationary AR(1) residual generator $\varepsilon_i(t) = \phi\,\varepsilon_i(t-1) + \eta_i(t)$ with i.i.d. innovations η_i tuned so that $\mathrm{Var}(\varepsilon_i) = \sigma_{\eta,i}^2/(1-\phi^2) = \sigma_{\varepsilon,i}^2$. The parameter ϕ leaves the joint distribution at a single time slice unchanged. It therefore does not change $\mathbb{E}(g_p)$ in Eqs. (S10) and (S13), but it changes the long-run variance of the sample mean ($\mathrm{Var}(\bar{g}_p) \to T^{-1}\,\sigma_p^2\,(1+\phi)/(1-\phi)$). The synthetic-ensemble median across 1,000 paths therefore has the same limit for i.i.d. and positively autocorrelated residuals. Serial dependence changes per-path sampling noise, not the ensemble median. The raw median's $\sim$22-percentage-point excess over an external long-run growth prior is therefore not caused by i.i.d. sampling *across time* at the level of the second-order formula.

The direct structural difference is contemporaneous. The derivation assumes residuals are orthogonal across assets at each time slice, as they are in the independent per-asset generator. Real residuals share sector and style factors, which reduce $\mathrm{Var}(\varepsilon_i - \varepsilon_j)$ below the independent-generator value $\sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$. Matching each marginal variance does not restore this covariance. The synthetic D_ε will therefore exceed its real-market counterpart when the omitted covariance is positive. Measuring that contribution requires a joint-residual experiment, which we did not run. The remaining excess includes higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals have non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons [54]. The location-shift correction of Eq. (S14) treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

To measure the bias, we applied a 20-asset, long-only target-weight allocator to the hybrid-composed universe. We calibrated the assets over 2014–2024, deployed the allocator in 2025, limited the universe to names with calibrated (α, β) and $\max\beta = 1.21$, and assumed a 5-basis-point round-trip cost. We then decomposed the path-mean annualized growth rate $\bar{g}_p \equiv (T\,\Delta t)^{-1}\,\log(W_{p,T}/W_{p,0})$

Table S2: **Diversification-return decomposition on a 20-asset target-weight allocator deployed against the hybrid-composed 2025 universe.** Median path-mean annualized growth rate $\bar{g}_p$ across the synthetic ensemble. Equal-weight buy-and-hold serves as a passive baseline; the target-weight allocation is reported under static (no intra-year rebalance) and daily-rebalanced regimes, with and without a signal on top of the target weights. The static-to-daily increase of $+27.73$ pp on the same target weights isolates the diversification return of Eq. (S13). The same allocator deployed against the real 2025 growth-rate history is shown for comparison.

Allocation rule (synthetic 2025 universe)	$\bar{g}_p$ **(%/yr)**
Equal-weight buy-and-hold	7.68
Target-weight, static (no rebalance)	1.97
Target-weight, daily rebalanced	29.69
Target-weight, daily rebalanced + signal	29.83
Real 2025 history, same allocator	11.50

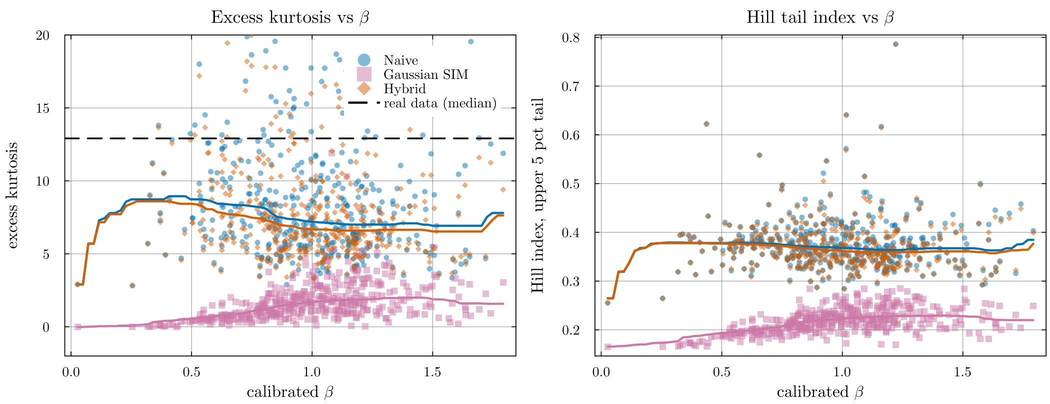

Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composition methods (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians remain near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22. The hybrid's median being below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (both retain the generator's tails) while Gaussian SIM does not.

into static and rebalancing-driven components (Table S2). The static-to-daily-rebalance switch increased the median by 27.73 percentage points. Adding a signal after daily rebalancing changed it by another 0.13 percentage points. The same allocator on the real 2025 growth-rate history produced $11.5\%/\text{yr}$, about 18 percentage points below the synthetic median. The synthetic residuals were independent across tickers and therefore omitted the contemporaneous covariance present in real markets. Determining how much of the gap this omission caused would require a joint-residual experiment with the marginals and allocator held fixed.

The correction subtracts a constant drift in log-wealth so that the ensemble median annualized growth rate matches an external prior $\bar{g}^{\text{prior}}$. At each time t, it multiplies every path by the same factor and therefore preserves cross-path wealth ranks. It also shifts every log-return increment by the same constant, which leaves the variance of those increments unchanged. For path p, the correction is

$$W_{p,t}^{\text{corr}} = W_{p,t} \cdot \exp\bigl(-b \cdot t\,\Delta t\bigr), \qquad b = \operatorname*{median}_{p} \bar{g}_p - \bar{g}^{\text{prior}}, \tag{S14}$$

with Δt the integration step in years and b, $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units. The correction is a linear detrend in log-wealth and remains multiplicative in wealth. It changes drawdown dates and depths, so it adjusts the return level rather than preserving drawdowns. It does preserve terminal-wealth ranks and the variance of log-return increments. The prior $\bar{g}^{\text{prior}}$ is set from

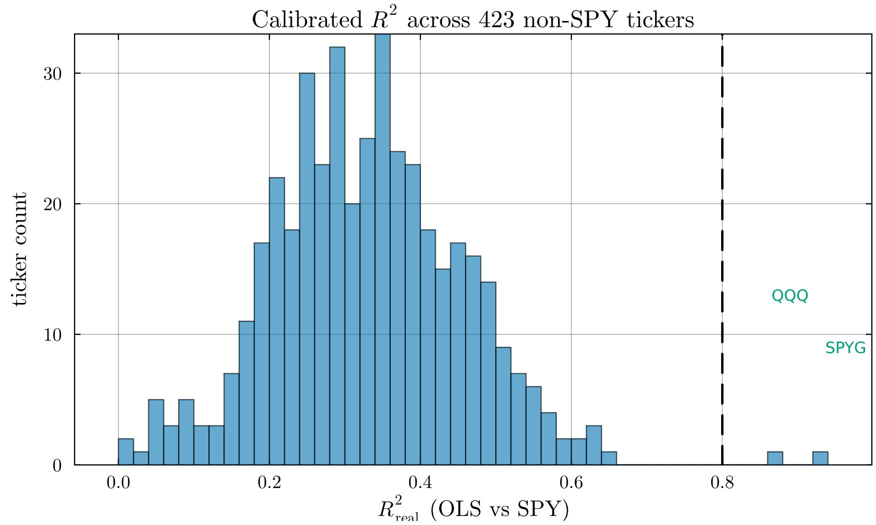

Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\mathrm{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) falls ~ 0.16 below the threshold. The R^2-preserving branch therefore contains only QQQ and SPYG in this universe.

external long-run data and is typically near $8\%/\mathrm{yr}$ at the time of writing: the prevailing risk-free rate (roughly $4.5\%/\mathrm{yr}$ on a five-year Treasury basis) plus the historical long-run equity premium of 4–6 pp [56, 57], less 1–2 pp for fees, turnover, and market-impact frictions, leaving ~ 3.5 pp of active premium net of costs. We used this documented prior rather than the realized test year to avoid circular calibration. Matching the realized year would force the synthetic median to equal the test sample. In this deployment, the real-2025 outcome of $11.5\%/\mathrm{yr}$ fell just above the median of the corrected ensemble.

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a tuning target and removes only a subset of the bias's consequences while leaving the underlying bias intact. Inflating per-trade transaction costs to absorb the diversification return would require round-trip costs of ~ 70 basis points versus the 1–10 basis-point range assumed for liquid US equity, shifting the misrepresentation from the synthetic generator to its cost model. Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, left the median Sharpe ratio essentially unchanged across the sweep, so standard-error-scale parameter drift did not absorb the bias. Imposing an AR(1) on the residual draw would conflict with the copula reorder step because reordering destroys temporal structure. Preserving both structures would require a block-ordered copula sampler and therefore a new residual model. The per-ticker block bootstrap [52] preserves each ticker's temporal residual structure but not covariance across tickers. A joint bootstrap or residual copula would be required for that purpose. The block bootstrap also replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the raw-versus-static deviation, $W^{\mathrm{corr}} = W_{\mathrm{static}} \cdot (W_{\mathrm{raw}}/W_{\mathrm{static}})^k$, hits the same target at k chosen to match the prior, but it changes the rank ordering of paths and inherits the static allocation's drawdown pattern.

The location-shift correction preserves percentile ranks and the relative ranking of strategies evaluated on the same paths. It also preserves centered moments of terminal log growth, including variance, skewness, and kurtosis. Because it changes the drawdown distribution, we evaluated drawdowns on the uncorrected ensemble. The correction does not repair residual autocorrelation, absolute Sharpe under daily rebalancing, or sensitivity to rebalancing frequency. Those targets require a residual model that preserves both temporal and cross-sectional dependence.

S3 Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov-Smirnov statistic between the empirical cumulative distribution functions (CDFs) of the observed and simulated growth-rate samples ($G_{\rm obs}$ of length n and $G_{\rm sim}$ of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ [42, 43]; the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson-Darling statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2 / [F(x)(1-F(x))]\, dF(x)$, where F is the pooled empirical CDF [44]. The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \left| G_{\rm obs}^{(i)} - G_{\rm sim}^{(i)} \right|, \tag{S15}$$

with $G^{(i)}$ the ith order statistic. The Hill upper-tail index is the classical order-statistic estimator:

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log\left(\frac{X_{(i)}}{X_{(k)}} \right), \tag{S16}$$

with $X_{(1)} \geq X_{(2)} \geq \dots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α, $\hat{\xi} \approx 1/\alpha$. The Hellinger distance between two density estimates p and q on a common bin grid is:

$$H(p, q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_2 \in [0, 1], \tag{S17}$$

with $H = 0$ for identical densities and $H = 1$ for disjoint supports.

S4 Transition matrix internals and partition diagnostics

Figure S5 shows the fitted HMM-WJ components for SPY with $N = 100$ states. The fitted Laplace CDF and the empirical CDF are shown with the 99 state boundaries Q_k. The transition matrix $\mathbf{T}$ has a near-diagonal band, with most off-diagonal mass within ± 5 states of the diagonal. Under the ordinary Markov transitions, every state has an expected residence time of only 1–2 steps. The fitted jump duration is $\lambda = 100$ steps, two orders of magnitude longer. Every state had at least one in-sample observation.

S5 Emission distribution and semi-Markov comparison

To compare the emission and duration specifications, we evaluated three variants on SPY (Table S3): the HSMM baseline of Bulla and Bulla [16] ($K = 8$ states, negative-binomial dwell times, Student-t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student-t emissions ($N = 100$). The HMM-WJ variants exceeded the HSMM's in-sample KS pass rate by 15 percentage points and its AD pass rate by 49 percentage points regardless of emission choice. The HMM-WJ variants use 100 states and the HSMM uses 8, so this comparison does not assign the pass-rate difference solely to the duration model. Holding the rest of HMM-WJ fixed, changing the emission from Normal to Student-t ($\nu = 5$) increased simulated excess kurtosis from 5.5 to 7.6, against an observed value of 7.7. Between the two HMM-WJ emissions, KS and AD pass rates changed by 0.6 and 0.8 percentage points, ACF-MAE and Wasserstein-1 were unchanged to three decimal places, and Hellinger changed by 0.001.

S6 State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for HMM-NJ and HMM-WJ. Pass rates varied little across $N \in \{60, \dots, 200\}$. At $N = 30$ both models lost several percentage points on the AD test. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined. We adopted $N = 100$ for all subsequent experiments.

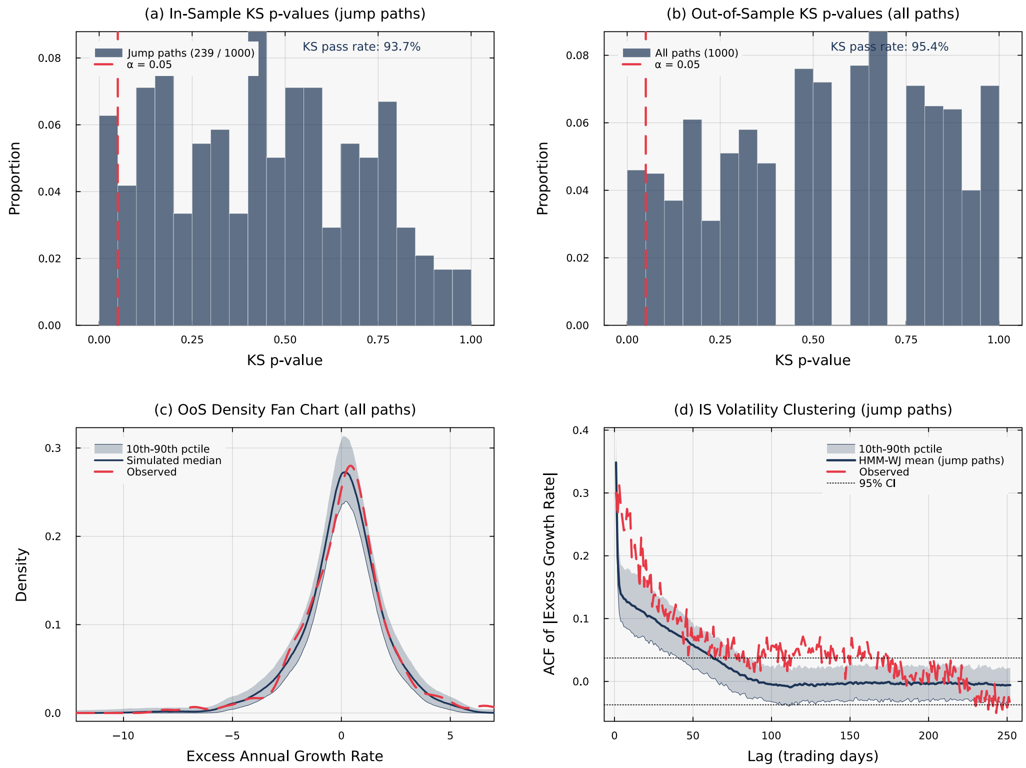

Figure S4: **Statistical validation of HMM-WJ** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p-values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p-values for all $1{,}000$ paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) Observed and mean simulated in-sample ACF of $|G_t|$ for jump paths.

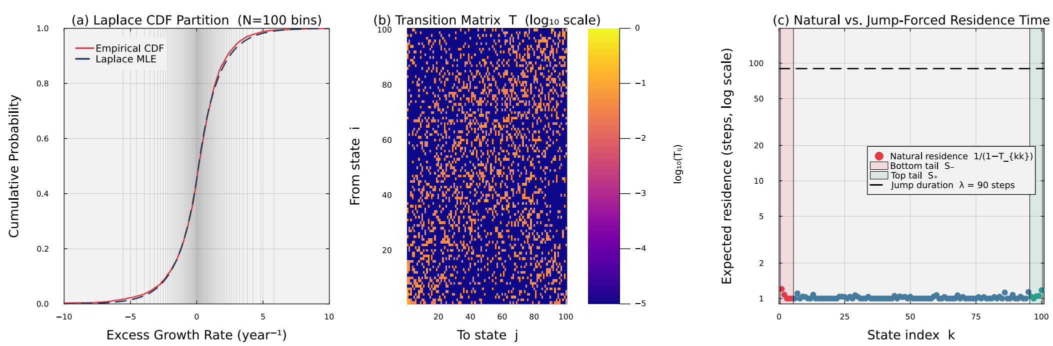

Figure S5: **Fitted HMM-WJ internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the fitted mean jump duration $\lambda = 100$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

Table S3: **Three-way comparison of emission distributions and duration specifications for SPY.** Standard errors in parentheses; bold marks the best value per row. HSMM: hidden semi-Markov model ($K = 8$ states, negative-binomial dwell times, Student-t emissions) [16]; HMM-WJ: hybrid hidden Markov model with jump-duration override ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$.

Metric	HSMM (Student-t)	HMM-WJ (Gaussian)	HMM-WJ (Student-t)
States	$K = 8$	$N = 100$	$N = 100$
Parameters	18	4	4
In-sample (2,766 trading days, 2014–2024)			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	**97.6** (0.5)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	**91.3** (0.9)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	**7.6** (0.14)
ACF-MAE	0.059 ($<$0.001)	**0.052** ($<$0.001)	**0.052** ($<$0.001)
Wasserstein-1	0.176 (0.001)	**0.101** (0.002)	**0.101** (0.001)
Hellinger	0.113 ($<$0.001)	0.076 ($<$0.001)	**0.075** ($<$0.001)
Out-of-sample (249 trading days, 2025)			
KS pass rate (%)	**96.2** (0.6)	95.0 (0.7)	94.4 (0.7)
AD pass rate (%)	**96.7** (0.6)	95.9 (0.6)	95.1 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	**6.1** (0.13)
ACF-MAE	0.042 ($<$0.001)	0.040 ($<$0.001)	**0.039** ($<$0.001)
Wasserstein-1	0.287 (0.002)	**0.275** (0.005)	0.282 (0.006)
Hellinger	0.239 (0.001)	0.212 (0.001)	**0.210** (0.001)

Table S4: **Hybrid composition method broken out by branch.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2-preserving branch (QQQ and SPYG) had median absolute errors of 0.005 for β and 0.001 for R^2, with a KS pass rate of 99%.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

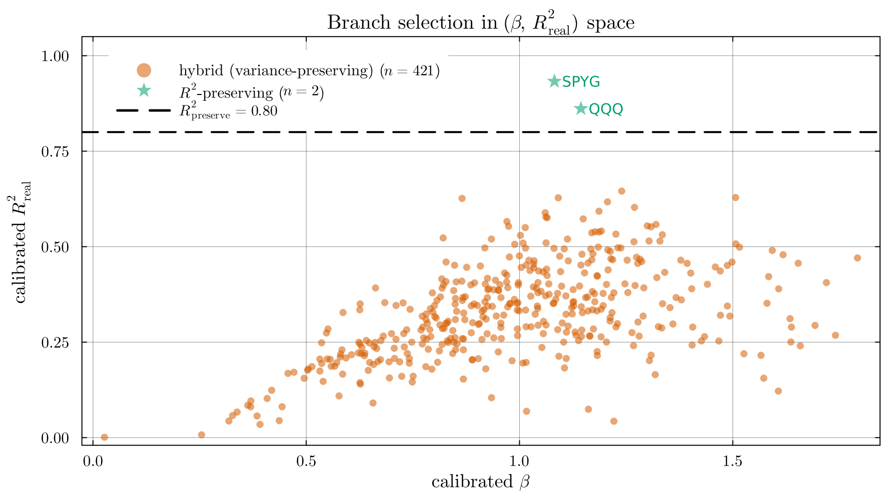

Figure S6: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2-preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

Table S5: **Aggregate metrics by β quartile, across composition methods.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows; the naive KS pass rate falls from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table S6: **Clipping-branch stress test under market-volatility scaling.** The market growth-rate path was rescaled by $\gamma \in \{1, 2, 3\}$ with the calibrated marginals held fixed, and the full hybrid composition method re-evaluated on the 423-ticker universe. As γ rises, the share of clipped tickers grows and the hybrid KS pass rate increases rather than falls.

γ	**Clipped (%)**	**Hybrid KS (%)**
1	0.0	50.3
2	76.4	71.8
3	95.2	77.3

S7 Cross-asset generalization

We repeated the fitting and evaluation procedure on three stocks: NVDA (semiconductors, high volatility), JNJ (consumer staples, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at $(n_{\mathrm{paths}} = 200, \mathrm{seed} = 1234)$ are reported in Table S7. The objective combines ACF and kurtosis errors with fixed weights. Because kurtosis varies across finite samples of heavy-tailed paths, the selected grid point can change with the Monte Carlo sample. The reported (ϵ^*, λ^*) values are reproducible with the stated path count and seed, but they are not unique per-ticker optima. At the selected parameter values, in-sample HMM-WJ KS pass rates were 98.9% for NVDA, 99.4% for JNJ, and 99.1% for JPM. Out-of-sample KS pass rates on the 249-day 2025 hold-out were 97.7%, 74.0%, and 79.2%, respectively.

S8 SIM composition derivations and properties verification

The variance-corrected single-index composition states two closed-form results, the variance scale of Eq. (7) and the residual-variance target of Eq. (10). The derivation below gives the large-T OLS coefficients and composed variance. To derive Eq. (7), set $\varepsilon_i = s_i \tilde{g}_i$ for $s_i \in (0, 1]$ and require that

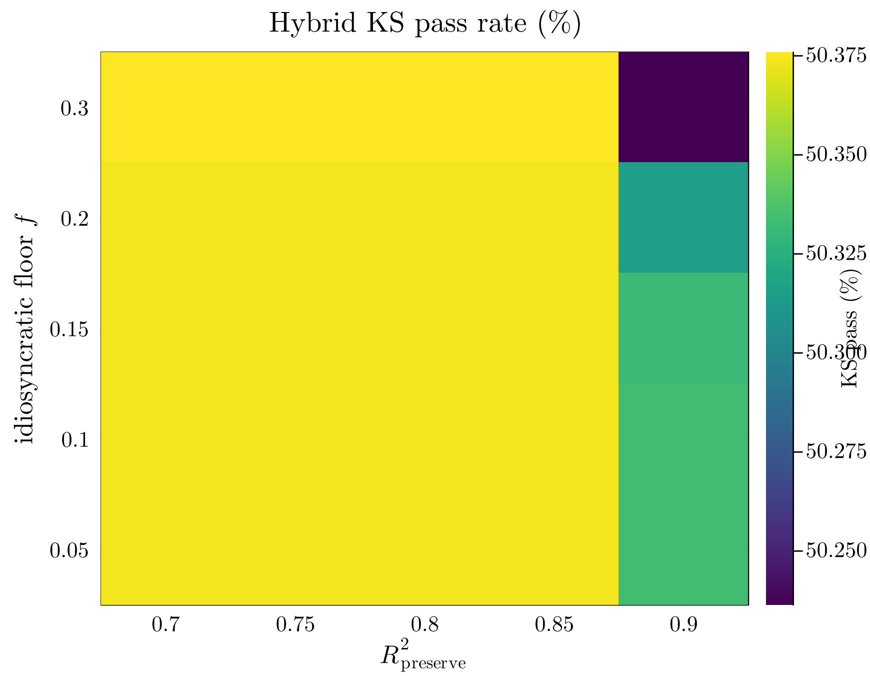

Figure S7: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The parameter values $(0.80, 0.10)$ used throughout the paper produced a KS pass rate within this range.

Table S7: **Cross-asset generalization of HMM-WJ across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed $= 1234$, and the corresponding HMM-WJ pass rates on 1,000 simulated paths at $\alpha = 0.05$ against each ticker's in-sample (2014–2024) and out-of-sample (2025) growth-rate history. Standard errors in parentheses (binomial SE for pass rates). The (ϵ^*, λ^*) values are reported as the parameter values used for the IS / OoS metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	**IS KS (%)**	**IS AD (%)**	**OoS KS (%)**	**OoS AD (%)**
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (consumer staples)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

Table S8: **Seed-uncertainty summary across 8 random seeds.** Each cell reports mean $\pm$ standard deviation (SD) across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker $\times$ 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The hybrid–JumpHMM-on-residuals difference in KS pass rate is 25 percentage points, compared with hybrid's across-seed SD of 0.18 percentage points.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000

the composed series have the same variance as the generator path:

$$\mathrm{Var}(g_i) \;=\; \mathrm{Var}(\alpha_i + \beta_i\, g_m + \varepsilon_i) \;=\; \mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \sigma^2_{\mathrm{gen},i}, \tag{S18}$$

where α_i drops out because it is constant. Expanding the sum gives

$$\mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \beta_i^2\; \mathrm{Var}(g_m) \;+\; 2\beta_i\; \mathrm{Cov}(g_m,\, \varepsilon_i) \;+\; \mathrm{Var}(\varepsilon_i). \tag{S19}$$

Substituting $\varepsilon_i = s_i\, \tilde{g}_i$ gives $\mathrm{Var}(\varepsilon_i) = s_i^2\, \sigma^2_{\mathrm{gen},i}$ and $\mathrm{Cov}(g_m,\, \varepsilon_i) = s_i\; \mathrm{Cov}(g_m,\, \tilde{g}_i)$. Under the assumption $\mathrm{Cov}(\tilde{g}_i,\, g_m) \approx 0$, the cross term vanishes and the variance condition becomes $\beta_i^2\, \sigma_m^2 + s_i^2\, \sigma^2_{\mathrm{gen},i} = \sigma^2_{\mathrm{gen},i}$. Solving for s_i^2 gives Eq. (7).

To derive Eq. (10), write the population R^2 identity for Eq. (4):

$$R^2 \;=\; \frac{\beta_i^2\, \sigma_m^2}{\beta_i^2\, \sigma_m^2 + \sigma^2_{\varepsilon,i}}, \tag{S20}$$

Solving this identity for the residual variance gives Eq. (10). In the large-T limit, OLS of g_i on g_m gives $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\mathrm{eff}}$. Here $\beta_i^{\mathrm{eff}} = \beta_i$ unless Eq. (9) clips the market loading. On the variance-preserving branch, the composed variance is $\sigma^2_{\mathrm{gen},i}$. Without clipping, $\mathrm{Var}(g_i) = \beta_i^2\sigma_m^2 + s_i^2\sigma^2_{\mathrm{gen},i} = \sigma^2_{\mathrm{gen},i}$. Under clipping, Eq. (9) sets β_i^{eff} to give the same variance. On the R^2-preserving branch, the composed variance is $\beta_i^2\sigma_m^2/R^2_{i,\mathrm{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\mathrm{real}})$ rather than by the generator marginal. The floor guarantees only that $\mathrm{Var}(\varepsilon_i) \geq f\, \sigma^2_{\mathrm{gen},i}$; it does not guarantee that adding the market term preserves the generator's tail distribution. In Table S5, the KS pass rate fell from 67% in the lowest-beta quartile to 44% in the highest-beta quartile.